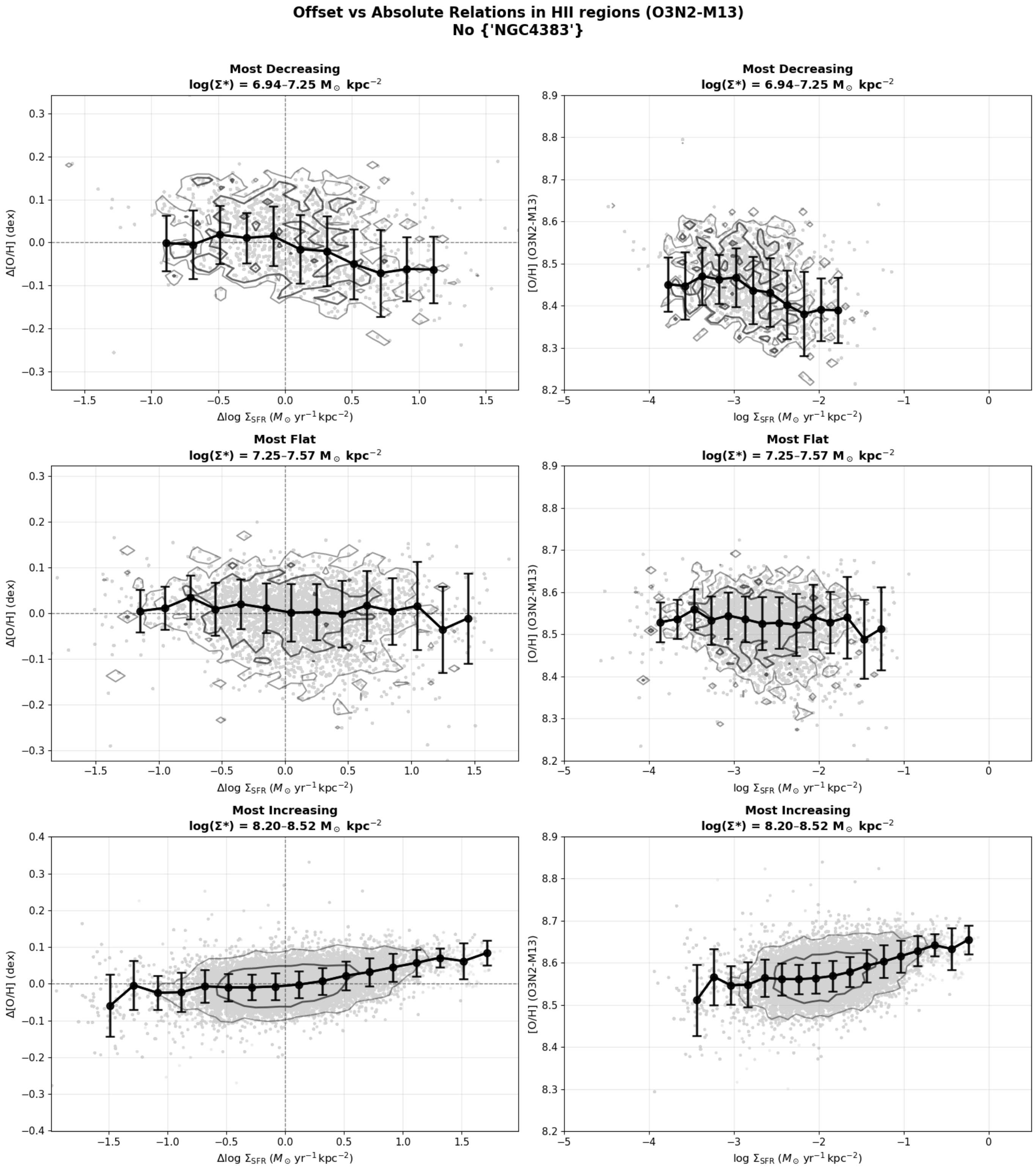
20251019 Offset and Absolute Relations in HII Regions

In short, the Σ_*-dependent increasing/decreasing trends in $[\mathrm{O/H}]$ vs Σ_{SFR} relation is real. The reason why we dont see it in the offset relation ($\Delta[\mathrm{O/H}]$ vs $\Delta\Sigma_{\mathrm{SFR}}$) is probably becuase they overlap each other when we putting each bin together.

For example, if you pick the most decreasing, flat and increaing Σ_* bins from the combined $[\mathrm{O/H}]$ vs Σ_{SFR} relation (right column), and create the associated $\Delta[\mathrm{O/H}]$ vs $\Delta\Sigma_{\mathrm{SFR}}$ relation (left column), then we can see that those offset relations are just renormalized in new parameter space and the trends are preserved.


Offset vs Absolute Relations in HII regions (O3N2-M13)
No {'NGC4383'}
Most Decreasing
log(Σ*) = 6.94–7.25 M☉ kpc⁻²
Most Flat
log(Σ*) = 7.25–7.57 M☉ kpc⁻²
Most Increasing
log(Σ*) = 8.20–8.52 M☉ kpc⁻²
Δ[O/H] (dex)
[O/H] (O3N2-M13)
Δlog Σ_SFR (M☉ yr⁻¹ kpc⁻²)
log Σ_SFR (M☉ yr⁻¹ kpc⁻²)